# Black-Peg Permutation Mastermind

Formal model with one extra coordinate check per round - definitions, logical constraints, and the bound objective

## 1 Definition of the game

Let **$[n] = \{1,\dots,n\}$** and let **$S_n$** be the set of permutations of [n]. The codemaker fixes an unknown secret permutation **pi in $S_n$**. The codebreaker learns pi through adaptive rounds.

**Black-peg query.** Submit $q_t$ in $S_n$ and receive the number of exact position matches

```
b_t = B_pi(q_t) = | { i in [n] : q_t(i) = pi(i) } |.
```

**Extra check.** Choose one pair $(i_t, j_t)$ in [n] x [n] and receive the truthful bit

```
c_t = C_pi(i_t,j_t) = 1 iff pi(i_t) = j_t; otherwise c_t = 0.
```

**Convention.** The strategy may choose $q_t$ from the earlier transcript, observe $b_t$, and then choose the extra check. If both questions must be fixed simultaneously, say so explicitly; that is a weaker model.

## 2 Equivalent logical constraints

Introduce Boolean variables $x_{ij}$, where $x_{ij}$ = 1 exactly when pi(i) = j. A transcript through round r is consistent precisely when all of the following hold:

| | |
|---|---|
| **Permutation** | `For every i: sum_j x_ij = 1; for every j: sum_i x_ij = 1; and x_ij in {0,1}.` |
| **Black pegs** | `For every t <= r: sum_i x_(i,q_t(i)) = b_t.` |
| **Extra checks** | `For every t <= r: x_(i_t,j_t) = c_t.` |

Let $V_r$ be the set of permutations satisfying these constraints. The game is solved exactly when **$|V_r| = 1$**; the remaining element is the secret pi.

## 3 Instruction and research goal

| INSTRUCTION | GOAL |
|---|---|
| For each n, construct an adaptive strategy that chooses permutation queries and targeted checks from the transcript. Prove that the constraints force a unique secret, and count the maximum number of rounds over all pi in $S_n$. | Let $T_+(n)$ be the minimum worst-case number of rounds. Find functions L(n) and U(n) such that **$L(n) <= T_+(n) <= U(n)$**. Make the bounds as tight as possible - ideally matching in order, or exactly. |

## 4 Immediate benchmark bounds

**Lower bound (decision tree).** A round has at most 2(n+1) possible answer pairs. Distinguishing all n! secrets therefore requires

```
[2(n+1)]^T >= n! => T_+(n) >= ceil( log(n!) / log(2(n+1)) ) = Omega(n).
```

**Upper bound (inheritance).** The extra check may be ignored, so every classical permutation black-peg strategy still works. In particular, **$T_+(n)$ = O(n log n)**. Thus the first target is to close the gap **Omega(n) versus O(n log n)**, using the targeted check in the upper-bound strategy or strengthening the adversary argument.

Reference for the standard black-peg AB/permutation model and its classical O(n log n) strategy: M. El Ouali, C. Glazik, V. Sauerland, and A. Srivastav, *On the Query Complexity of Black-Peg AB-Mastermind*, arXiv:1611.05907. The extra-check model and the decision-tree lower bound are stated here explicitly.